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Automatic detection of keratoconus on Pentacam images using feature selection based on deep learning

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Abstract

Today, corneal refraction, height, and thickness data, which are required in the diagnosis of keratoconus, can be obtained with corneal tomography devices. Pentacam four map display presenting this data is one of the most basic options in the diagnosis of keratoconus. In this article, an artificial intelligence-based method using Pentacam images is proposed to distinguish keratoconus from healthy eyes. Axial/sagittal curvature, back elevation, front elevation, and corneal thickness map images of a total of 341 keratoconus and 341 healthy corneas obtained from İnönü University ophthalmology clinic as the data set were given as input to AlexNet, one of the deep learning models, and the feature vectors of each image were obtained and combined. The most effective features in the determination of keratoconus were determined by applying ReliefF, minimum-redundancy-maximum-relevance (mRMR) and Laplacian algorithms, which are widely used in feature extraction algorithms, to the obtained feature vector. These features are classified using the support vector machine (SVM) classifier, which has high performance in binary classification. The accuracy, specificity, and sensitivity of keratoconus detection with the proposed method were found to be 98.53%, 99.01%, and 98.06%, respectively. The developed model can support the clinician to evaluate the features of the cornea and to detect keratoconus, which is difficult through subjective assessments, especially in the subclinical and early stages of the disease.

KEYWORDS

deep learning, feature selection, keratoconus, Pentacam four maps refractive

1 | INTRODUCTION

Keratoconus is a corneal ectasia characterized by progressive corneal thinning and protrusion resulting in irregular astigmatism and vision loss.^{1,2} The disease usually starts in adolescence and can progress up to the fourth decade.³ Although various factors initiate the disease and are effective in its progression, its etiology has not been fully elucidated.⁴ Subclinical keratoconus

can only be differentiated by corneal imaging methods and no visual impairment can be found clinically. It is very difficult to treat optic and visual problems that occur with the progression of the disease to advanced stages when there is no visual impairment in the subclinical period.⁵ Therefore, corneal tomography is essential for the diagnosis of the disease in the early stages, the detection of risk factors for progression, and the decision of corneal crosslinking treatment.^{5–8} In

addition, the detection of the subclinical form before refractive surgeries is of great importance in terms of preventing iatrogenic corneal ectasia.⁹

Pentacam (Oculus Optikgeräte GmbH, Wetzlar, Germany) is an anterior segment tomography device that can obtain corneal thickness, front and back elevation maps thanks to its rotating Scheimpflug camera. The color-coded four map display is a Pentacam display where corneal curvature, thickness, front elevation, and back elevation maps can be viewed together. Keratoconus can be diagnosed in the presence of abnormal patterns specific to each map on the color-coded four map display with various numerical cut-off values and indices. However, in corneas with subclinical keratoconus, a normal pattern can often be observed in the four map display. In other words, pathological change in subclinical keratoconus may not appear in the color-coded four map. Diagnostic screens and analyzes have been added to Pentacam, such as the Belin/Ambrósio Enhanced Ectasia screen, to distinguish cases with subclinical keratoconus from healthy corneas.¹⁰ However, learning the use of these screens imposes a knowledge burden on the clinician. In addition, the presence of the auxiliary screen other than the four map makes simultaneous evaluation difficult and causes a waste of time. For these reasons, a practical method is required for accurate and rapid diagnosis, especially in subclinical keratoconus.

While technology and especially artificial intelligence are entering many areas of life day by day, important developments are experienced in the field of health. Artificial intelligent (AI) applications in healthcare are becoming more common, especially to support clinicians' diagnostic decisions and automate certain tasks. AI has features that can improve health fields such as ophthalmology within the framework of a new vision. Artificial intelligence algorithms have been used successfully in the evaluation of eye-related conditions such as glaucoma,¹¹ retinal diseases,¹² and cataracts.¹³ In recent years, deep learning-based studies that support the clinician's decision in the detection of keratoconus disease have attracted attention.^{14–16}

Kuo et al. used three different convolutional neural network (CNN) models for keratoconus screening based on corneal topographic images. Sensitivity and specificity were determined at 0.90 levels with these models. Using the VGG16 model, they revealed that ophthalmologists focused on the largest gradient difference of topographic maps corresponding to diagnostic clues.¹⁶ Kamiya et al. evaluated the diagnostic accuracy of keratoconus using deep learning of color-coded maps measured by swept-source anterior segment optical coherence tomography (AS-OCT). Using deep learning, the six color-coded maps

(front elevation, front curvature, back elevation, back curvature, total refractive power, and pachymetry map) had an accuracy of 0.991 in distinguishing between healthy and keratoconus eyes. The proposed deep learning model showed an accuracy of 0.874 in classifying the stage of the disease.¹⁷ Shi et al. proposed a neural network model based on corneal morphological features to distinguish eyes with subclinical keratoconus from healthy eyes from UHR-OCT image data. As a result of the classification of the feature vectors extracted from each image, the area under the ROC curves (AUC) was determined as 0.93.¹⁸ Lavric et al. developed a machine learning algorithm that can detect keratoconus in its early stages. Proposed machine learning algorithms to detect keratoconus were tested with medical data, including corneal topography, elevation, and pachymetry parameters collected from OCT-based topography devices. The highest accuracy of 94% was achieved through the SVM algorithm.¹⁹ Shetty et al. in their retrospective study performed the local and global evaluation of keratoconus with three different AI methods with a total of 1884 Pentacam images of 366 eyes (296 patients). Accuracy for each model was 90%, 91%, and 93%. The proposed models identified eyes with concordant progression-related changes in $K_{\max}$ and other parameters.²⁰ Abdelmotaal et al. proposed a CNN architect for the diagnosis of keratoconus with images obtained from color-coded corneal maps by Pentacam HR. The architecture has 13 layers and the accuracy value was 95.8%. The proposed model showed less successful results in the detection of keratoconus in the front elevation map, and the highest accuracy and sensitivity were found in the thickness map.¹⁴ To detect keratoconus, Chen et al. proposed a CNN model that classified each of the axial, thickness, front elevation and back elevation maps of Pentacam HR four map images separately. The obtained accuracy was between 92.83% and 97.49% for each map. The accuracy of the classification performed by combining the images of the four maps was 97.85%.²¹ Feng et al. classified the raw data of front axial, back axial, thickness, front height, and back height maps consisting of numerical matrix in Pentacam HR with KerNet model containing multilevel fusion structure. They reported accuracy of 94.15% in data with subclinical keratoconus and 97.08% in data with keratoconus.²² Table 1 summarizes the various studies on the automatic classification of keratoconus.

Table 1 presents the classification results with the latest techniques. Our main motivation is to detect keratoconus from Pentacam four map images and improve classification accuracy. For this purpose, we propose a feature selection-based deep learning model to differentiate keratoconus from healthy eyes. The

TABLE 1 Artificial intelligence techniques developed for keratoconus detection

Ref.	Algorithm	Instruments	Input	Number of classes	Data	Performance
Kuo et al., 2020	VGG16 Resnet152 InceptionV3	Pentacam	Corneal Topographic Images	3	170 KC, 28 Sub-KC, 156 Healthy	AUC: 90%
Kamiya et al., 2019	Resnet-18	Pentacam	optical coherence tomography (AS-OCT)	2	304 KC, 239 Healthy	A:[82–99.1]%
Shi et al., 2020	Neural network	Pentacam HR	Metrics obtained in the Pentacam HR system	3	50 Healthy, 38 KC, 33 Sub-KC	AUC:93%
Lavric et al., 2020	SVM	OCT-based topography	Corneal Parameters	3	1970 Healthy 791 forme fruste KC, 390 KC	A:94%
Shetty et al., 2021	Three Model based on AI	Pentacam	Corneal parameters	2	1884 Healthy +KC	A:93%
Abdelmotaal et al., 2020	CNN	Pentacam HR	color-coded Scheimpflug images	3	1038 KC, 1108 Healthy, 1072 Sub-KC	A:95.8%
Chen et al., 2021	CNN	Pentacam HR	axial, front elevation, back elevation and pachymetry	2	134 Healthy, 1926 KC	A:97.85%
Feng et al., 2021	KerNET	Pentacam HR	Numerical Data	3	335 Healthy, 172 Sub-KC, 347 KC	A:97.08

innovations and contributions of the proposed method are given below.

- Convolutional neural networks are the best models for solving computer vision problems. However, they have high computational complexity. Hand-made feature generation models have a low time load, but hand-modeled learning methods cannot extract high-level features and cannot provide high performance on large image datasets. Therefore, in this article, deep learning-based architecture is proposed to distinguish Keratoconus from healthy eyes in Pentacam images. In architecture, axial/sagittal curvature, back elevation, front elevation, and corneal thickness images are evaluated together.
- The feature vectors from each image were combined. In this feature vector, the features with the highest classification accuracy were selected. For this, three different feature selection algorithms were used.
- Subclinical data participated in the evaluation.
- It is difficult to make subjective evaluations, especially in the subclinical and early stages of the disease. The proposed method to overcome this challenge is a helpful tool for physicians in assessing the corneal conditions and detecting keratoconus.

2 | DATA

This retrospective cross-sectional study was performed with Pentacam images of patients who applied to the Malatya İnönü University ophthalmology department between May 2010 and July 2021. A total of 682 pentacam four map images obtained. Three hundred forty one eyes of 178 unilateral or bilateral keratoconus patients and 341 eyes of 178 healthy controls were included in the study.

Figure 1 shows an example Pentacam four map screen images. Images of patients with corneal disease other than keratoconus and other ocular or systemic conditions that may affect the results were not included in the study. This study followed the tenets outlined in the Declaration of Helsinki. Institutional Review Board (IRB)/Ethics Committee approval was obtained from Malatya İnönü University Health Sciences Non-Invasive Clinical Research Ethics Committee.

Color-coded four-map display of Pentacam presents the sagittal curvature, front elevation, back elevation, and thickness map with the location of their numeric markers. The diagnosis of keratoconus was made by evaluating Pentacam findings in addition to decreased vision and slit-lamp findings related to keratoconus obtained from patient files (M.F.).¹ Only images reporting OK in

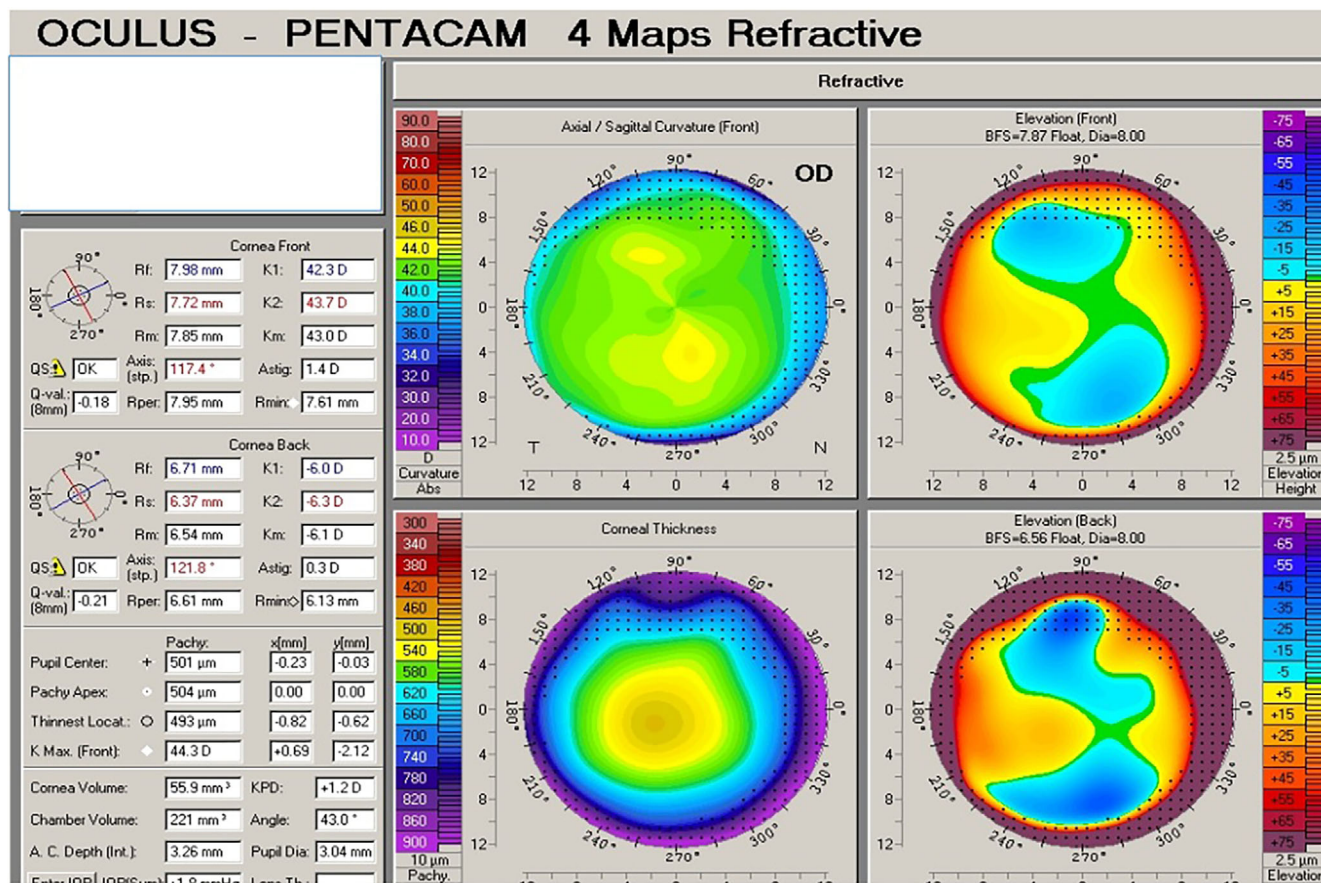


FIGURE 1 Example Pentacam four map screenshot

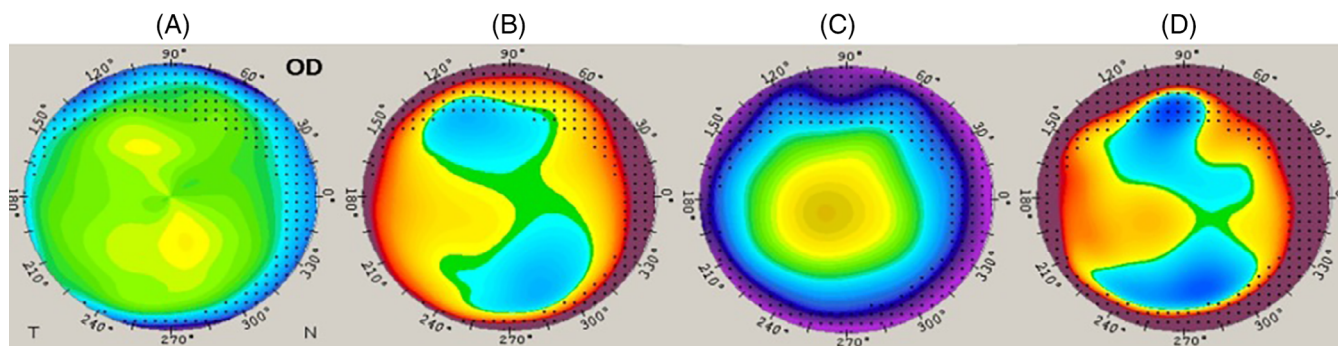


FIGURE 2 Map of sagittal curvature (A), front height (B), thickness (C), and back height (D)

the quality score section were included in the study. Images were acquired with the standard 25 scan option. Subclinical keratoconus was accepted as mild corneal tomographic changes without slit-lamp findings suggestive of keratoconus or changes in visual acuity. Subclinical keratoconus was not taken as a separate group and was included in the keratoconus group.

In addition to the standard color-coded four map in JPG format with numerical and spatial markers removed, age, gender, eye laterality, flattest keratometry

curvature (K_1), steepest keratometry curvature (K_2), mean keratometry curvature (K_m), maximum curvature power (K_{max}), thinnest corneal thickness (TCT), corneal astigmatism values were recorded. Each four map image was divided into four maps of $227 \times 227 \times 3$ pixels (Figure 2).

SPSS version 25.0 (IBM corp., Chicago, USA) was used for statistical evaluation of age, gender, eye laterality, K_1 , K_2 , K_m , K_{max} , corneal astigmatism, TCT data. Continuous variables were presented as mean

$\pm$ standard deviation (SD), while categorical data were presented as frequency (n) and percentage (%). Continuous variables were evaluated with the Independent samples T -test, and categorical data were evaluated with the Pearson Chi-square test. A p -value less than 0.05 was considered statistically significant.

3 | THE PROPOSED SYSTEM

In order to reveal the characteristic findings of keratoconus disease in the very early stages, tomographic parameters are limited and there may be difficulties in its diagnosis. We propose a method that evaluates together the axial/sagittal curvature, back elevation, front elevation, and corneal thickness map images obtained from Pentacam in order to take into account all the details that cannot be noticed in the clinician's evaluation. Our presented method has three

basic stages and they are feature extraction, feature selection, and classification.

First, axial/sagittal curvature, front elevation, back elevation, and corneal thickness map images of each patient are given as input to the AlexNet model, and feature vectors are obtained. Second, these feature vectors are combined. At this stage, the feature selection algorithm is applied to select the most distinctive and effective features. Thus, the features that maximize the classification success are determined. Finally, classification is performed using the feature vector. A schematic of the proposed deep learning-based model for differentiating keratoconus and healthy eyes using Pentacam images is shown in Figure 3.

The general stages of the presented model are described below. The first stage of the model is the acquisition of axial/sagittal curvature, front elevation, back elevation, and corneal thickness map images from Pentacam images.

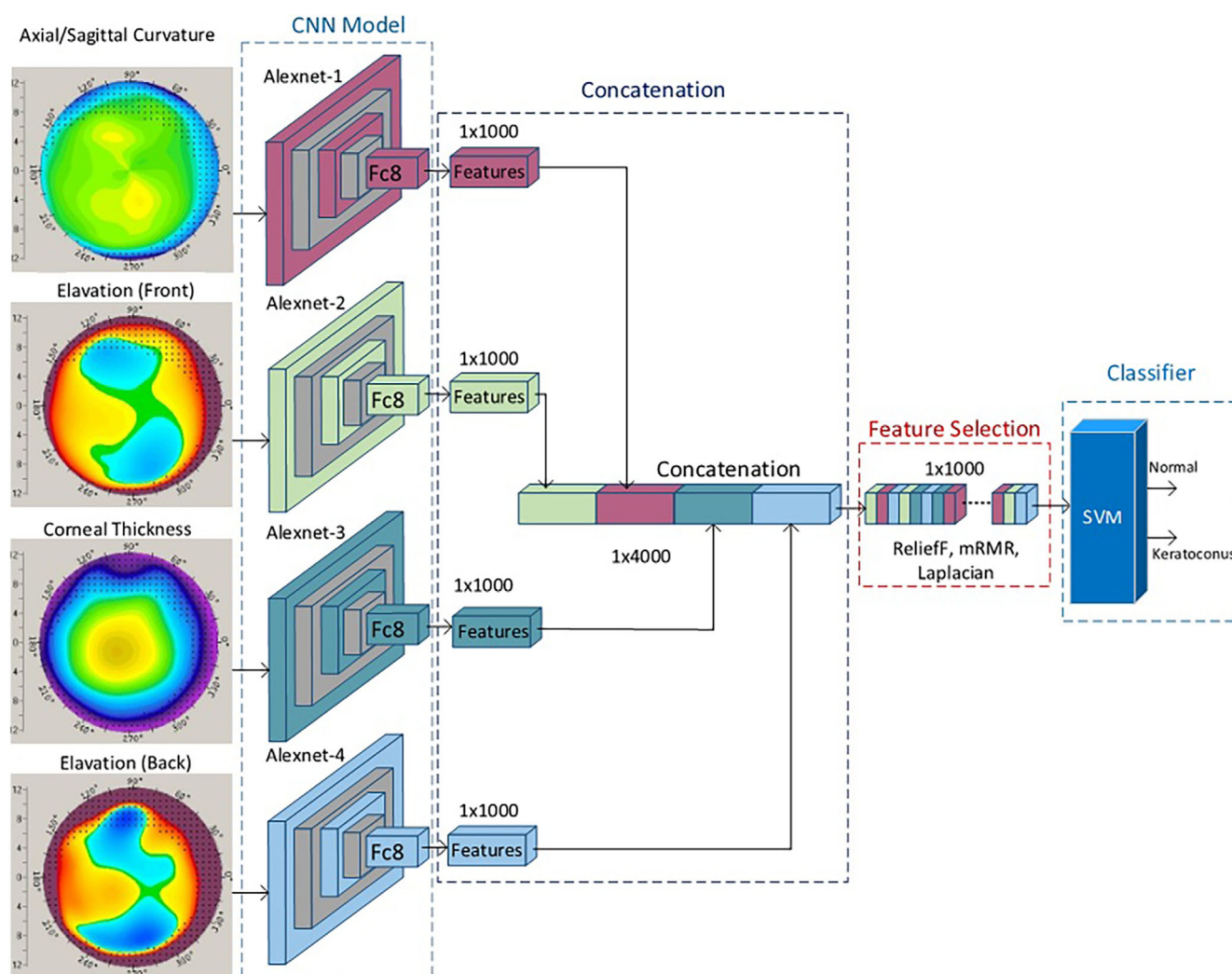


FIGURE 3 The framework of the proposed method

3.1 | Step 0: Feed axial/sagittal curvature, front elevation, back elevation, and corneal thickness map images into AlexNet architectures

AlexNet is the first architecture to popularize convolutional neural network models and deep learning again in 2012. It is a convolutional neural network architecture designed by Krizhevsky et al.²³ In this architecture, there are 25 layers in total, including 1 data, 5 convolutional, 7 Relu (Rectified Linear Unit), 2 normalizations, 3 max-poolings, 3 fully connected, 2 dropouts, 1 softmax, and 1 output. The data layer used as input is $227 \times 227 \times 3$ images. When categorizing images with CNNs, feature extraction is done first. In the AlexNet architecture, features can be obtained from layers such as pooling or fully connected.

3.2 | Step 1: Extract feature vectors in the fc8 layer of AlexNet architectures

In the proposed architecture, features in the fully connected (fc8) layer, which is the 23rd layer of the AlexNet architecture, are used. In this layer, a feature of 1×1000 size is obtained for each image. The reason for using the features in the fc8 layer is that it is the layer before the classifier (softmax) layer. Thus, the feature vectors of the fc8 layer, which are the most decisive, are used for classification. Subsequently, the features from the four images are concatenated to obtain a single large feature vector.

3.3 | Step 2: Concatenate feature vectors

Feature vectors obtaining from step 1 are concatenated to obtain a new feature vector. The concatenated feature vector is 1×4000 in size and is the features of a patient's axial/sagittal curvature, back elevation, front elevation, and corneal thickness images. The mathematical definition for the concatenate function is as follows:

$$\mathbf{F}_{1fc8} = \text{AlexNet}(\mathbf{I}_A), // \mathbf{I}_A : \text{Axial/Sagittal Curvature Image}$$

$$\mathbf{F}_{2fc8} = \text{AlexNet}(\mathbf{I}_B), // \mathbf{I}_B : \text{Back Elevation Image}$$

$$\mathbf{F}_{3fc8} = \text{AlexNet}(\mathbf{I}_C), // \mathbf{I}_C : \text{Front Elevation Image}$$

$$\mathbf{F}_{4fc8} = \text{AlexNet}(\mathbf{I}_D), // \mathbf{I}_D : \text{Corneal Thickness Image}$$

$$\mathbf{F} = \text{Concat}(\mathbf{F}_{1fc8}, \mathbf{F}_{2fc8}, \mathbf{F}_{3fc8}, \mathbf{F}_{4fc8})$$

The Concat function combines the feature vectors of all four AlexNet models to create a new feature vector of 1×4000 size.

3.4 | Step 3: Apply a feature selection algorithm to determine the features that maximize the classification success in the obtained feature vector

The feature vector of size 1×4000 obtained at the end of the second step is given to the feature selection algorithms and ordered according to their rank values. In this study, ReliefF, mRMR, and Laplacian feature selection algorithms were used to determine the rank of feature vectors.

3.4.1 | ReliefF Feature Selection Algorithm

This algorithm tries to determine the values of the features and whether there is a dependency between the features. To achieve this result, it compares the closest samples in the classes to which the samples belong and are not weighted. This method first selects a sample from the data set. Secondly, it creates a model according to the proximity of the relevant example with other examples in its own class and its distance from different classes and then performs the feature selection process. The ReliefF formula is given as in Equation (1).

$$W_i = W_{i-1} - (x_i - \text{nearHit}_i)^2 + (x_i - \text{nearMiss}_i)^2 \quad (1)$$

W_i indicates the weight (significance) of the i th feature, nearHit_i the relevant feature value in the closest sample with the same class, and nearMiss_i the related feature value in the closest sample with a different class.

3.4.2 | Minimum Redundancy Maximum Relevance (mRMR) Algorithm

The mRMR algorithm finds an optimal set of features that are mutually and maximally different and can effectively represent the response variable.²⁴ The algorithm minimizes the redundancy of a feature set and maximizes the relationship of a feature set to the response variable. The purpose of this algorithm is to find a set of F features that make the V_s value given in Equation (2) maximum and the W_s value given in Equation (3).

$$V_s = \frac{1}{|F|} \sum_{x \in F} I(x, y) \quad (2)$$

$$W_s = \frac{1}{|F|^2} \sum_{x, z \in F} I(x, z) \quad (3)$$

Here, I is mutual information value and the ratio of V_s to W_s is MIQ (mutual information quotient) value. It determines the feature that makes the MIQ value the largest at Equation (4) F_c .

$$\max_{x \in F^c} MIQ_x = \max_{x \in F^c} \frac{I(x, y)}{\frac{1}{|F|} \sum_{z \in F} I(x, z)} \quad (4)$$

3.4.3 | Laplacian score

LS is based on the observation that two data points that are close to each other are probably related to the same class.²⁵ For any data point, the distance (D_{ij}) is calculated using the nearest neighbor method. The distance is converted into the similarity matrix (Equation (5)) using the kernel function.

$$S_{ij} = \exp\left(-\left(\frac{D_{ij}}{\sigma}\right)^2\right) \quad (5)$$

where σ denotes the scale factor for the kernel function.

The Laplacian score is determined with the help of the equation by calculating the mean ($\tilde{x}_r$) and score (S_r) for each feature.

$$\tilde{x}_r = x_r - \frac{x_r^T D_g \mathbf{1}}{\mathbf{1}^T D_g \mathbf{1}} \quad (6)$$

where x_r is the r th feature, D_g is the degree matrix, and $\mathbf{1}^T = [1, \dots, 1]^T$.

$$S_r = \frac{\tilde{x}_r^T S \tilde{x}_r}{\tilde{x}_r^T D_g \tilde{x}_r} \quad (7)$$

$$L_r = 1 - \frac{\tilde{x}_r^T S \tilde{x}_r}{\tilde{x}_r^T D_g \tilde{x}_r} \quad (8)$$

L is the Laplacian matrix, defined as the difference between D_g and S .

3.5 | Step 4: Identify the features that maximize the classification success from the feature vectors ordered according to their rank and obtain the classification results

The first 1×1000 feature vector from the 1×4000 feature vector ordered according to their rank in step three is selected for classification. In this process, classification is made using sequential permutations of the 1×1000 feature vector. For this purpose, the SVM classification

algorithm is used. SVM, one of the machine learning algorithms, is widely used in binary classification problems and provides high success. In the SVM algorithm, the c parameter is selected in the range of $[10^{-3}$ to $10^3]$ and the gaussian radial basis function is used as kernel function. The reason for this is that according to the literature, it is known that high success accuracy can be achieved especially in binary classification. The feature vectors are given to the SVM input as ordered permutations and the classification success is recorded.

3.6 | Step 5: Calculate confusion matrix and performance parameter

The confusion matrix is used and performance parameters are calculated to determine the success of the proposed model in distinguishing patients with keratoconus. The pseudo-code of the process applied to distinguish keratoconus patients from healthy ones with the proposed method is as follows (Algorithm 1):

Algorithm Pseudo code of proposed method

Input: Pentacam Image I_p (I_A , I_F , I_C , I_B)

Output: Healthy or Keratoconus

// Feature extraction of images from AlexNet's

$F_{1fc8} = \text{AlexNet1}(I_A)$, // I_A : Axial/Sagittal Curvature Image

$F_{2fc8} = \text{AlexNet2}(I_F)$, // I_F : Back Elevation Image

$F_{3fc8} = \text{AlexNet3}(I_C)$, // I_C : Front Elevation Image

$F_{4fc8} = \text{AlexNet4}(I_B)$, // I_B : Corneal Thickness Image

// Concatenate features

$F = \text{Concat}(F_{1fc8}, F_{2fc8}, F_{3fc8}, F_{4fc8})$

// Determine the score or rank value of the obtained 1×4000 dimensional feature vectors

$F_s = \text{ReliefF}(F)$ or $F_s = \text{mRMR}(F)$ or

$F_s = \text{Laplacian}(F)$

// Determine the feature that maximizes the classification accuracy (F_{SVM})

for $i:1$ to 1000

for $j:1$ to i

$F_{SVM} = F_s(1, j)$

 SVM (F_{SVM} , **class**) // Classify the new feature vector using the SVM classification algorithm.

end

end

4 | EXPERIMENTAL RESULTS

In this study, there was no significant difference between the control and keratoconus groups in terms of gender, eye laterality distribution, and mean age. Mean K_1 , K_2 , K_m , $K_{\max}$, corneal astigmatism were significantly greater in the keratoconus group than in the control group ($p < 0.001$ for all). The mean TCT was significantly thicker in the control group than in the keratoconus group ($p < 0.001$) (Table 2).

A total of 75 (21.99%) corneas in the keratoconus group had subclinical keratoconus. Mean K_1 , K_2 , K_m , $K_{\max}$, and TCT in patients with subclinical keratoconus were 43.33 ± 1.38 D, 45.45 ± 2.11 D, 44.35 ± 1.47 D, 48.31 ± 2.50 D, and 496.03 ± 28.56 μ m, respectively. In eyes with subclinical keratoconus, mean K_1 , K_2 , K_m , $K_{\max}$ were significantly steeper ($p < 0.001$ for all), and mean TCT was significantly thinner ($p < 0.001$) compared to the control group. According to the Amsler-Krumeich classification, 151 (44.28%) eyes were grade 1, 68 (19.94%) eyes were grade 2, 11 (3.22%) eyes were grade 3 and 36 (10.55%) were eye grade 4.

In order to objectively evaluate the classification success of the proposed method, first of all, axial/sagittal curvature, front elevation, back elevation, and thickness map images were classified separately with AlexNet architecture. Of the 642 Pentacam images, 70% were used for training and 30% for testing. The resulting classification accuracy was below 93.63% for each map image, and the accuracy values were 91.67%, 90.69%, 85.29%, and 93.63% for each AlexNet architecture, respectively (Table 2).

The method proposed in this article evaluates axial/sagittal curvature, back elevation, front elevation, and thickness map images together. ReliefF, mRMR, and

Laplacian feature selection methods were applied to determine the feature vector that maximizes the classification accuracy. Feature vectors obtained by ReliefF, mRMR and Laplacian methods were ordered according to their rank values. When the first 141 features obtained by the ReliefF method and ordered according to their rank values, SVM performs the classification with maximum accuracy. The obtained accuracy value is 98.53%. In the classification performed by using the first 301 features obtained by the mRMR method and ordered according to their rank values, the accuracy value is 98.53%. The maximum accuracy obtained when using 341 features with the Laplacian method with the lowest accuracy value was 93.14%. Figure 4 shows the variation in classification accuracy obtained using sequential permutations of the first 1000 features.

The confusion matrix was used to evaluate the performance of the method. The confusion matrix is used to comprehensively evaluate the developed model. Eleven performance evaluation metrics were used to calculate the ability of the method to identify patients with keratoconus, including accuracy (A), sensitivity (SN), specificity (SP), precision (P), F1-score (F1), misclassification rate (M), negative predictive value (NPV), false positive rate (FPR), false discovery rate (FDR), false negative rate (FNR), and diagnostic odds ratio (DOR). The number of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) were used to calculate these parameters. The mathematical definitions of these six parameters can be found in the following equations.

$$A = \frac{TP + TN}{TP + FN + TN + FP} \quad (9)$$

TABLE 2 Demographic characteristics and Pentacam findings of the control and keratoconus groups

	Control	Keratoconus	<i>p</i>
Age (mean $\pm$ SD)	25.82 $\pm$ 12.87	24.64 $\pm$ 7.48	0.145*
Gender (M)	97 (54.5%)	104 (58.8%)	0.418**
Eye (R)	177 (52.1%)	170 (49.9%)	0.565**
K_1 (mean $\pm$ SD)	42.31 $\pm$ 1.70	45.96 $\pm$ 4.25	<0.001*
K_2 (mean $\pm$ SD)	44.39 $\pm$ 2.11	49.37 $\pm$ 5.61	<0.001*
K_m (mean $\pm$ SD)	43.31 $\pm$ 1.77	47.58 $\pm$ 4.79	<0.001*
$K_{\max}$ (mean $\pm$ SD)	45.18 $\pm$ 2.13	54.09 $\pm$ 7.28	<0.001*
Astig. (mean $\pm$ SD)	2.08 $\pm$ 1.39	3.44 $\pm$ 2.41	<0.001*
TCT (mean $\pm$ SD)	529.92 $\pm$ 37.99	453.77 $\pm$ 48.23	<0.001*

Abbreviations: Astig., korneal astigmatizma; K_1 , flattest keratometry curvature; K_2 , steepest keratometry curvature; K_m , mean keratometry curvature; $K_{\max}$, maximum curvature power; M, male; R, right eye; SD, standard deviation; TCT, thinnest corneal thickness.

*Independent samples *T*-test.

**Pearson Chi-square test.

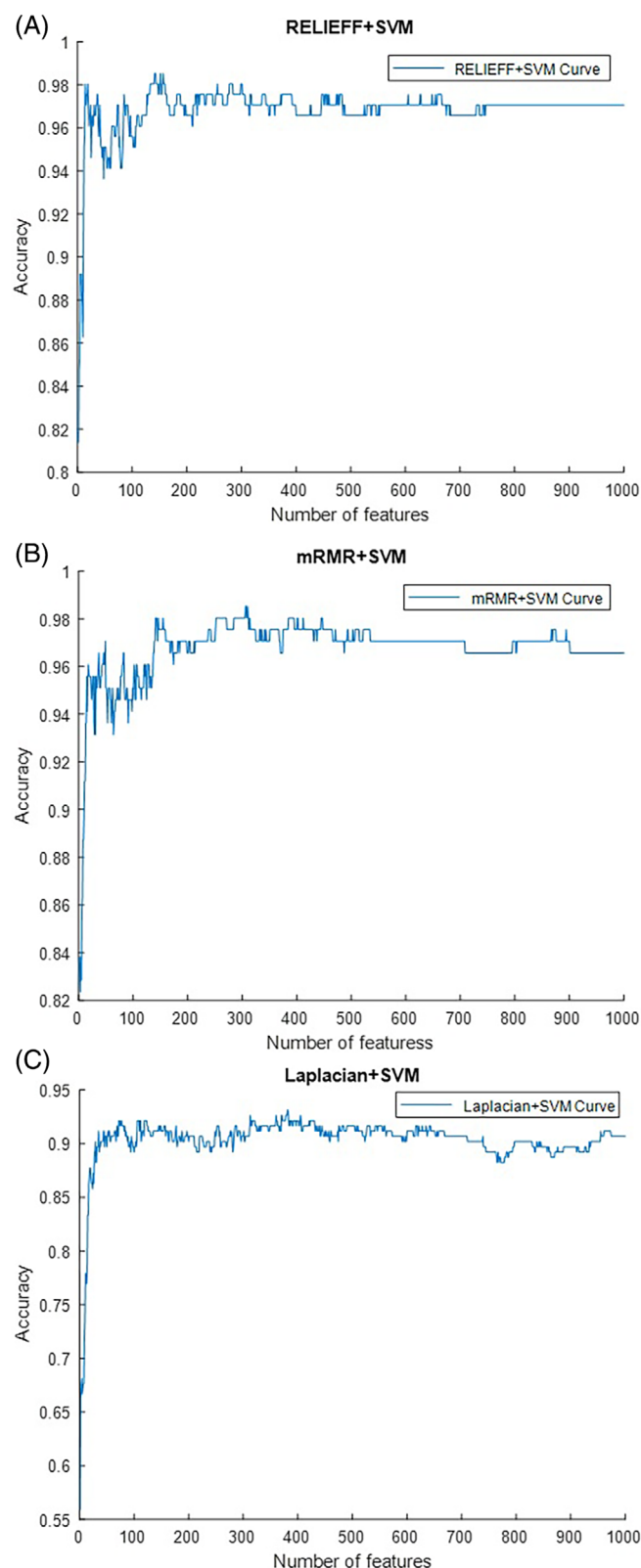


FIGURE 4 Accuracy changes obtained by selection of feature vectors used for classification (A) ReliefF, (B) mRMR, and (C) Laplacian

$$SN = \frac{TP}{TP + FN} \quad (10)$$

$$SP = \frac{TN}{TN + FP} \quad (11)$$

$$P = \frac{TP}{TP + FP} \quad (12)$$

$$F1 = \frac{2 \times SN \times P}{SN + P} \quad (13)$$

$$\text{Misclassification rate} = \frac{FP + FN}{TP + FN + TN + FP} \quad (14)$$

$$NPV = \frac{TN}{TN + FN} \quad (15)$$

$$FPR = \frac{FP}{FP + TN} \quad (16)$$

$$FDR = \frac{FP}{FP + TP} \quad (17)$$

$$FNR = \frac{FN}{FN + TP} \quad (18)$$

$$DOR = \frac{TP \cdot TN}{FP \cdot FN} \quad (19)$$

The confusion matrix obtained for the ReliefF and mRMR feature selection algorithm is the same. Therefore, the obtained performance parameters are the same. The lowest classification accuracy was obtained by using the Laplacian feature selection algorithm. Figure 5 shows the confusion matrices obtained by using the ReliefF, mRMR, and Laplacian feature selection algorithms used in the proposed method. Table 3 shows the values of the obtained performance parameters.

5 | DISCUSSION

Advanced keratoconus treatment costs are high and optimal results are often not achieved. Despite corneal cross-linking treatment, keratoconus is still among the top five causes of corneal transplantation. Early-stage keratoconus is still the biggest concern before refractive surgeries.^{26,27} For these reasons, it is very important for ophthalmology to detect keratoconus in the early stages

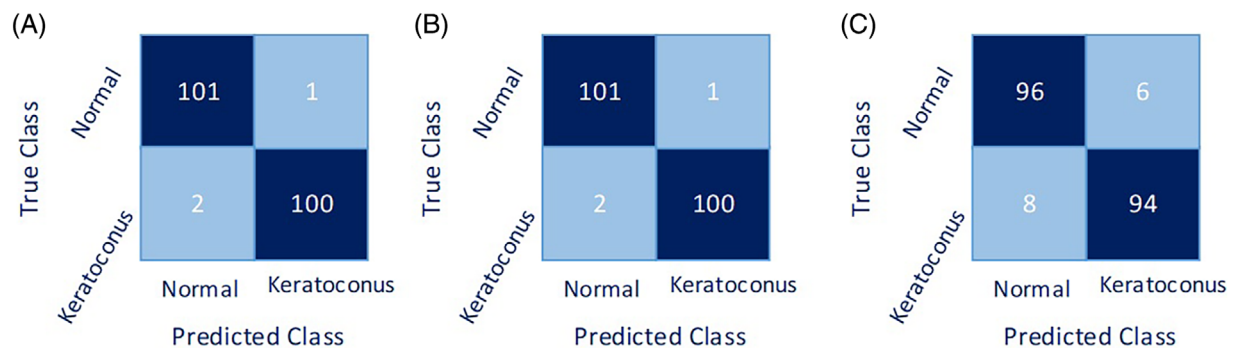


FIGURE 5 Confusion matrices (A) ReliefF, (B) mRMR, and (C) Laplacian

TABLE 3 Classification performance parameter values

Metrics	AlexNet ₁	AlexNet ₂	AlexNet ₃	AlexNet ₄	Proposed Method		
					ReliefF	mRMR	Laplacian
A (%)	91.67	90.69	85.29	93.63	98.53	98.53	93.14
SN (%)	92.08	91.92	84.62	92.38	98.06	98.06	92.31
SP (%)	91.26	89.52	86.00	94.95	99.01	99.01	94.00
P (%)	91.18	89.22	86.27	95.10	99.01	99.01	94.12
F1 (%)	91.63	90.55	85.44	93.72	98.53	98.53	93.20
M (%)	8.33	9.31	14.71	6.37	1.47	1.47	6.86
NPV (%)	91.18	92.16	86.27	95.10	98.04	98.04	92.16
FPR (%)	7.92	10.48	15.38	7.62	0.99	0.99	6.0
FDR (%)	7.84	10.78	15.69	7.84	0.98	0.98	5.88
FNR (%)	8.74	8.08	14.0	5.05	1.94	1.94	7.69
DOR	121.41	97.2	33.78	227.95	5050.0	5050.0	188.0

and take precautions. The back elevation map is a Pentacam map frequently used by the clinician in the diagnosis of keratoconus.²⁸ In our study, the highest accuracy, sensitivity, and specificity of the AlexNet architectures, in which the four map image was used separately, was obtained from the back elevation map (Table 2). This result shows parallelism with the importance of back elevation map in the clinical practice of ophthalmology. The second highest accuracy, sensitivity and specificity were obtained from the axial/sagittal curvature map. This result may be due to the reflection of keratometric changes on the map and the ability of artificial intelligence to precisely evaluate the color values and placements corresponding to the keratometric values. However, the success of the thickness map, in which the location of the thinnest point, its thickness value, and its relationship to its surroundings can be evaluated, was lower than other maps in the classification phase.²⁹

Eyes with keratoconus were detected with 98.53% accuracy using the proposed model to further increase the classification success achieved. This success was higher than the success of the results obtained by evaluating the four-map images separately.

In classification problems with two class labels, the diagnostic odds ratio (DOR) is a measure of the effectiveness of the diagnostic test. A DOR value greater than 1 indicates that the classification performance is high and successful. As can be seen from the results presented in Table 3, the DOR parameter value obtained with the proposed method is higher than the DOR value obtained with the pretrained AlexNet. Considering Table 1 comparatively, there are popular and up-to-date architectures among CNN architectures. The architectures used in the studies are pre-trained. With these architectures, it has been determined whether there is keratoconus on Pentacam images. In our proposed model, four different images in the

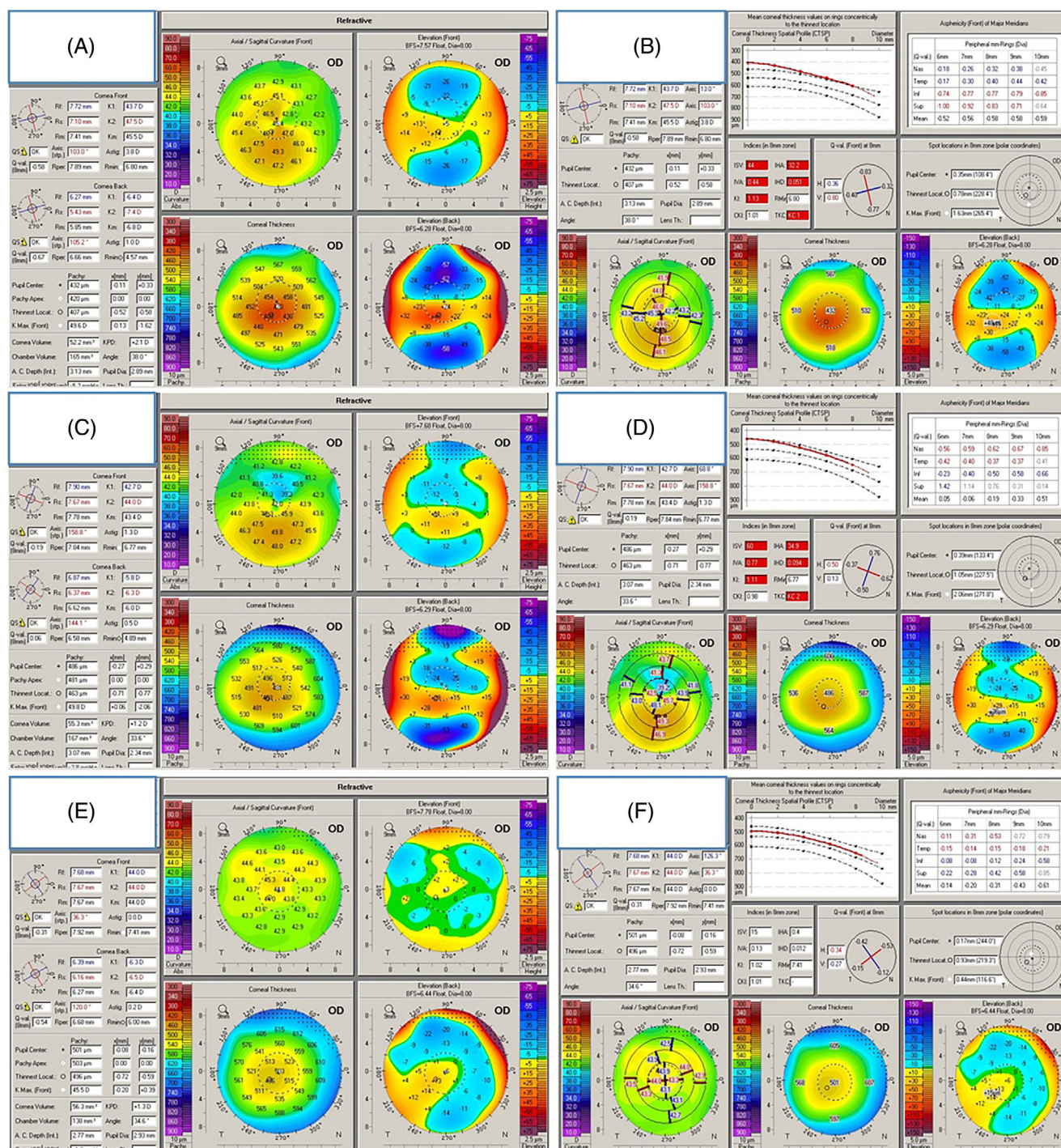


FIGURE 6 Misclassified four map images. In both four-map images [(A) and (C)] with keratoconus classified as healthy, paracentral cone location on both elevation maps and suspicious inferior displacement of the thinnest point in the thickness map and keratoconus findings on the topometric map are remarkable [(B) and (D)]. A junctional pattern (A) and inferior steep pattern (C) are seen on the sagittal curvature maps. The healthy case classified as keratoconus is a keratoconus suspect case with superior steepness in sagittal curvature map, tongue-like extension in back elevation map, and suspicious displacement of the thinnest point to the inferior (E). However, topometric map data are within normal limits

Pentacam image were evaluated. Separate feature vectors from these images were obtained with AlexNet architecture, feature vectors were combined and feature extraction was done. Feature selection was applied

to the feature vector and classification was made. The increase in success when the four maps were evaluated together demonstrates the importance of making decisions by integrating the information from the four

maps. In proposed model, all maps are evaluated together in order to increase the classification success as in clinical practice.

Our model had misclassified three images from total 204 four-map images (1.47%). Two images were classified as healthy (false negative) despite having Amsler-Krumeich stage 1 keratoconus. In contrast, there was one image from a control group that was classified as keratoconus (false positive) even though it was healthy (Figure 6).

The fact that our model misclassified one eye of two patients in the healthy group could be due to the relatively early stage of the cases. In addition, the eye that was misclassified as keratoconus, is a difficult case to decide because the patterns in the four-map may be pathologic, and data outside the four-map may be necessary for decision making. Our model correctly classified the other eyes of the same patients with similar map images in all three misclassified cases. This indicates that the classification error was due to not giving the model which eyes belonged to the same patient. Considering that cases that are difficult to diagnose in clinical practice are decided with additional clinical information such as age, visual acuity, refractive status, fellow eye status, apart from quadratic map information, it can be said that our model is quite successful without all these data. In our study, 226 (66.27%) of the cases with keratoconus were early-stage keratoconus. Considering that those with early-stage keratoconus constitute the majority of eyes included in the study, it can be said that our model successfully classifies early-stage keratoconus.

In their study of 61 healthy eyes and 20 eyes with subclinical keratoconus, Castro-Luna et al. classified subclinical keratoconus using the random forest model with 89% accuracy, 93% specificity, and 86% sensitivity.¹⁵ The progression of subclinical cases to clinically observable keratoconus suggests that subclinical cases have a similar but milder corneal disorder to those with keratoconus. For this reason, it would not be wrong to say that subclinical cases in the four map have changes similar to those of keratoconus, but cannot be distinguished by the human eye. Classification of subclinical cases in the same group as those with keratoconus may have positively affected the learning process of our model by increasing the content of the learning dataset thanks to the similarity in the maps. This might have enabled our model to classify subclinical cases with a high accuracy rate.

In addition to the advantages mentioned above, the proposed method also has some limitations:

- Larger databases may be collected in the near future and our model may be tested on larger databases of Pentacam images collected.

- It is likely to increase the classification accuracy by using the current and latest architectures in the literature instead of the AlexNet architecture used in the model.
- This model has not been tested on images of individuals with keratoconus and different corneal diseases.

6 | CONCLUSION

In conclusion, our study identified keratoconus disease from Pentacam images with high accuracy. Our method, in which four map images are evaluated together, can also successfully detect subclinical keratoconus. The algorithm can be integrated into corneal imaging devices or used as standalone software for corneal assessment and detection of early-stage keratoconus. Thus, it can be used for keratoconus screening before refractive surgeries due to its high reliability. In addition, thanks to the low cost and high data processing capacity of the model, it can be used in large surveys in areas where the disease is common. The proposed model can assist the clinician in assessing the difficult corneal condition and detecting keratoconus through subjective assessments, especially in the subclinical and early stages of the disease, and may prevent the clinician from dealing with a large number of mixed data and maps in corneal tomography.

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CONFLICT OF INTEREST

All authors read and approved the final manuscript. The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data available on request from the authors

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